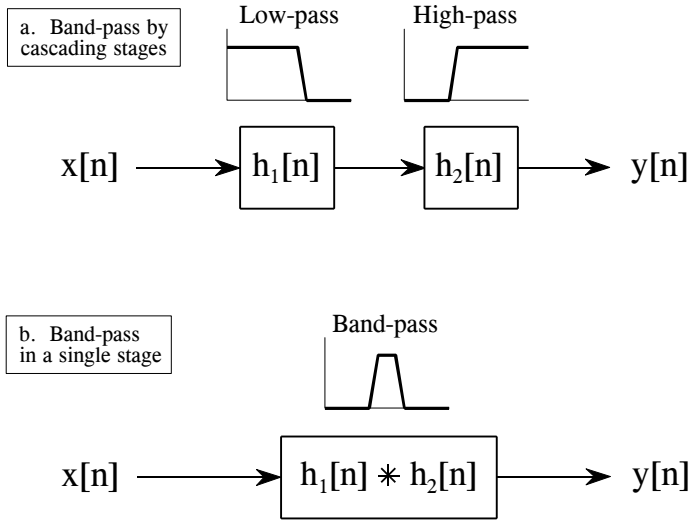

FIGURE 14-8
Designing a band-pass filter. As shown in (a), a band-pass filter can be formed by cascading a low-pass filter and a high-pass filter. This can be reduced to a single stage, shown in (b). The filter kernel of the single stage is equal to the *convolution* of the low-pass and high-pass filter kernels.

becomes 0. The cutoff frequency of the example low-pass filter is 0.15, resulting in the cutoff frequency of the high-pass filter being 0.35.

Changing the sign of every other sample is equivalent to multiplying the filter kernel by a sinusoid with a frequency of 0.5. As discussed in Chapter 10, this has the effect of *shifting* the frequency domain by 0.5. Look at (b) and imagine the negative frequencies between -0.5 and 0 that are of mirror image of the frequencies between 0 and 0.5. The frequencies that appear in (d) are the negative frequencies from (b) shifted by 0.5.

Lastly, Figs. 14-8 and 14-9 show how low-pass and high-pass filter kernels can be combined to form band-pass and band-reject filters. In short, *adding* the filter kernels produces a *band-reject* filter, while *convolving* the filter kernels produces a *band-pass* filter. These are based on the way cascaded and parallel systems are be combined, as discussed in Chapter 7. Multiple combination of these techniques can also be used. For instance, a band-pass filter can be designed by adding the two filter kernels to form a stop-pass filter, and then use *spectral inversion* or *spectral reversal* as previously described. All these techniques work very well with few surprises.

Filter Classification

Table 14-1 summarizes how digital filters are classified by their *use* and by their *implementation*. The use of a digital filter can be broken into three categories: *time domain*, *frequency domain* and *custom*. As previously described, time domain filters are used when the information is encoded in the shape of the signal's waveform. Time domain filtering is used for such actions as: smoothing, DC removal, waveform shaping, etc. In contrast, frequency domain filters are used when the information is contained in the